



Taxi Trip Price Prediction

Estimating fair taxi fares with Machine Learning
and a real-time, Google Maps-style location experience

Polynomial Regression • Streamlit • LocationIQ

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The Problem



Fares feel unpredictable

Riders rarely know the price before booking - distance, traffic, time of day and weather all move the number.



Manual entry is error-prone

Typing distance and duration by hand is slow, and easy to get wrong - especially on an unfamiliar route.



Location search feels clunky

Most course-project demos stop at a static map. Real riders expect live, Google Maps-style search.



₹ ? ? ?

"How much will this ride actually cost?"

- Every rider, every time

Project Objective

Build a Streamlit app that predicts taxi fares and feels like a real ride-hailing product



Predict Fare Accurately

Use a trained Polynomial Regression model to estimate Trip Price from real trip conditions.



Make Location Selection Effortless

Google Maps-style autocomplete and an interactive map replace manual lat/long entry.



Use Real Road Distance

Pull actual road distance & duration from a routing API instead of guessing, with a documented fallback.

Dataset Overview

taxi_trip_pricing.csv



Features used by the model

Trip_Distance_km	Time_of_Day	Day_of_Week
Passenger_Count	Traffic_Conditions	Weather
Base_Fare	Per_Km_Rate	Per_Minute_Rate
Trip_Duration_Minutes	Trip_Price (target)	

Tech Stack

Everything used to build, serve, and map this project



Python

Core language



Pandas

Data loading & cleaning



scikit-learn

Model training pipeline



Streamlit

Web application framework



Folium

Interactive map rendering



streamlit-searchbox

Live autocomplete UI



LocationIQ API

Geocoding & routing



Requests

HTTP calls to the API

Data Preprocessing Pipeline

Every prediction (and this deck's numbers) follows this exact order - it must match training exactly



Model artifacts (label_encoder, scaler, poly_features, model, feature_columns) are stored together in trip_price_model.pkl and reused as-is - never retrained at inference time.

The Model: Polynomial Regression

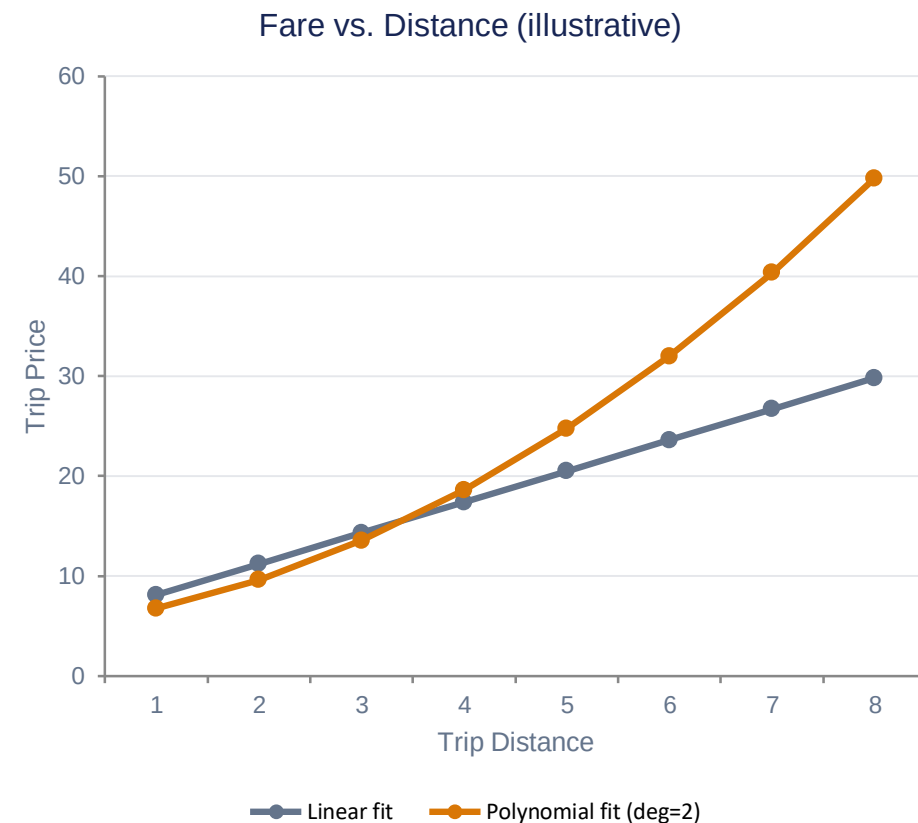


Degree = 2, fit with scikit-learn's LinearRegression

Why polynomial, not plain linear regression?

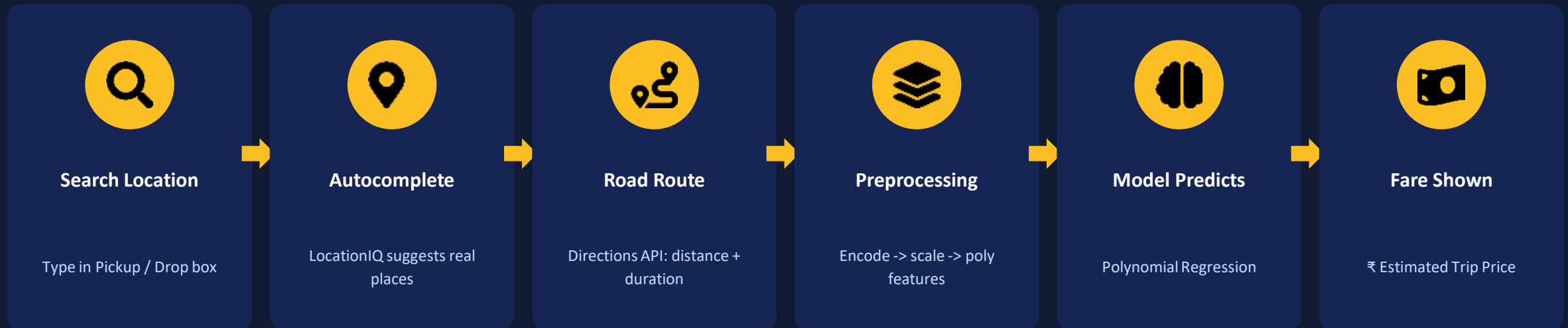
Fare doesn't grow in a perfectly straight line with distance or duration - short trips carry a fixed pickup cost, while long trips compound distance and time charges together.

PolynomialFeatures(degree=2) expands every feature (and its pairwise interactions) into a richer set, then an ordinary LinearRegression is fit on top - giving it room to capture that curve and cross-effects like Distance $\times$ Traffic.



How a Prediction Happens

End-to-end flow from typing a location to seeing a fare



If routing fails, the app falls back to straight-line (Haversine) distance - always clearly labelled, never silently swapped in.

Key Features



Live Autocomplete Search

Type 3+ characters and real place suggestions appear instantly, just like Google Maps - no typing raw coordinates.



Interactive Map & Route

Pickup and drop markers, plus the actual road route drawn on the map, auto-framed to fit both points.



Auto Distance & Duration

Road distance and estimated travel time are pulled live and pre-filled - editable if you want to explore a scenario.



Instant Fare Prediction

One click runs the full preprocessing + Polynomial Regression pipeline and shows the estimated price.

Maps & Routing: LocationIQ

An OpenStreetMap-based provider, properly licensed for autocomplete use



Autocomplete API

Live place suggestions as the user types



Reverse Geocoding

Turns a map click into a real address



Directions API

Road distance, duration & route geometry

Caching (st.cache_data) avoids re-calling the API for a query or route already looked up in this session.

5,000

free requests / day

No credit card required to start - more than enough for demo and portfolio use.

If routing is unavailable: falls back to Haversine (straight-line) distance - clearly labelled in the UI, never presented as road distance.

Challenges & Solutions

Challenge	Solution
Styled HTML rendered as literal code text in the UI	Stripped leading whitespace per line before <code>st.markdown()</code> , since Markdown treats indented lines as code blocks
No live suggestions while typing (only on Enter/blur)	Switched to <code>streamlit-searchbox</code> for true debounced, per-keystroke autocomplete
Public Nominatim isn't licensed for autocomplete use	Moved to LocationIQ - a properly licensed OSM-based provider with a free tier
Keeping the map, search box and clicks all in sync	Session-state-driven design: one source of truth for pickup/drop, widgets keyed off it

Future Scope

Where this project could go next



Cloud Deployment

Host on Streamlit Community Cloud or a small VPS for public access



Live Traffic Data

Swap static traffic levels for a real-time traffic-aware routing signal



Larger, Real-world Dataset

Retrain on real trip logs across more cities for better generalization



Fare Feedback Loop

Let users log actual fares paid to continuously validate predictions



Thank You

Questions & Discussion Welcome

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